

Lecture 2 - Induction and Cardinality

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Theorem 1. *The set $\wp(\mathbb{Q})$ is uncountable.*

Proof. We begin by restating the corollary found in the previous lecture that $\mathbb{Q}$ is denumerable. Since $\mathbb{Q}$ is denumerable, we can construct an ordered sequence of all the elements of $\mathbb{Q}$ in the form $(r_1, r_2, \dots, r_i, \dots)$. Let $S = \{s(s_i) \mid s_i \in \{0, 1\}\}$ represent the set of all sequences of zeros and ones. Letting $E \subseteq \mathbb{Q}$, consider the function:

$$f(E) = \begin{cases} 1, & \text{if } r_i \in E \\ 0, & \text{if } r_i \notin E \end{cases}$$

We can see that this function is injective, because every sequence in S has some corresponding subset in $\mathbb{Q}$ and surjective, since two different subsets of $E_1, E_2 \subseteq \mathbb{Q}$, $E_1 \neq E_2$ must have at least a single element not in common between them, and so necessarily produce different sequences. Therefore, f is a bijective function with a domain of all the subsets of $\mathbb{Q}$ ($\wp(\mathbb{Q})$) and a co-domain of all possible sequences of zeros and ones. Thus, we know that $\wp(\mathbb{Q})$ and S must have the same size. Now we know that if we can show that S is uncountable, then we have also successfully shown $\wp(\mathbb{Q})$ is uncountable.

To show S is uncountable, we proceed with an argument from contradiction. Assume, for the purposes of contradiction that S is indeed denumerable. Then, just as we did with $\mathbb{Q}$, we can construct an ordered sequence of all elements of S . But, then we can represent S with the following table.

$s \in S$	s_1	s_2	s_3	$\dots$	s_i	$\dots$
$s^{(1)}$	s_1^1	s_2^1	s_3^1	$\dots$	s_i^1	$\dots$
$s^{(2)}$	s_1^2	s_2^2	s_3^2	$\dots$	s_i^2	$\dots$
$s^{(3)}$	s_1^3	s_2^3	s_3^3	$\dots$	s_i^3	$\dots$
$\dots$	$\dots$	$\dots$	$\dots$	$\ddots$	$\dots$	$\dots$
$s^{(i)}$	s_1^i	s_2^i	s_3^i	$\dots$	s_i^i	$\dots$
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\ddots$

Now consider creating a sequence (s_k) along the diagonal of the table, where for each entry on the diagonal, we simply choose the opposite of whatever its value is (for example, if $s_1^1 = 0$, $s_k^1 = 1$ and vice versa). We can formalize this by defining the sequence as:

$$(s_k) = (s_k^i = 1 - s_i^i)$$

We therefore have a sequence (s_k) which consists of only zeros and ones (meaning it belongs to S) but differs from each entry in the table by exactly one value. This means that our table, which was made under the assumption that we were able to represent ALL elements in the set S (because we assumed S to be denumerable) does not contain (s_k) , which as stated earlier is in S .

Thus, by contradiction, S cannot be denumerable, and since we already determined that S and $\wp(\mathbb{Q})$ are of the same size, $\wp(\mathbb{Q})$ cannot be denumerable and is thus uncountable. $\square$

- An operator, such as $+$, is commutative if for any two elements of a space a, b : $a + b = b + a$
- An operator, such as $+$, is associative if for any three elements of a space a, b, c : $(a + b) + c = a + (b + c)$
- A field $\mathbb{F}$ is a set which possesses certain additive, multiplicative, and distributive properties given below:
 - Additive properties:
 1. An addition operator $(+)$ which is associative and commutative.
 2. An additive identity element (0) such that when added to an element $a \in \mathbb{F}$, $a + 0 = a$.
 3. All elements in $a \in \mathbb{F}$ have an additive inverse $-a \in \mathbb{F}$ such that $a + -a = 0$.
 - Multiplicative properties:
 1. A multiplication operator $(\cdot)$ which is associative and commutative.
 2. A multiplicative identity (1) such than when added to an element $a \in \mathbb{F}$, $a \cdot 1 = a$.
 3. All elements in $a \in \mathbb{F}$ have a multiplicative inverse (besides the additive identity) $\frac{1}{a} \in \mathbb{F}$ such that $a \cdot \frac{1}{a} = 1$.
 - Distributive properties:
 1. For all elements $a, b, c \in \mathbb{F}$, multiplication is distributive over addition $a \cdot (b + c) = a \cdot b + a \cdot c$.

Axiom 1. $\mathbb{R}$ is a field.

- This can actually be proven, but for the purpose of this course, this is assumed to be true.